

CS & IT Engineering



**Lecture Number
07**



By- Ankit Doyla Sir



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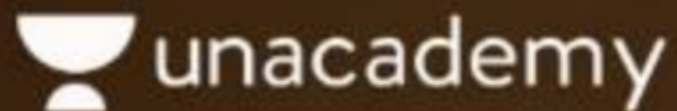
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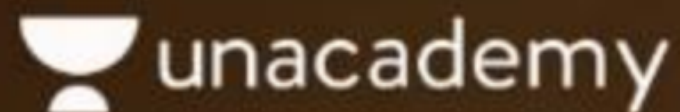


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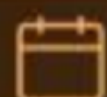
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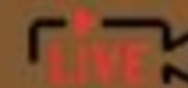


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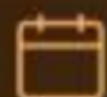


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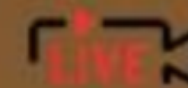


CSIT



Batch A

Live



Question

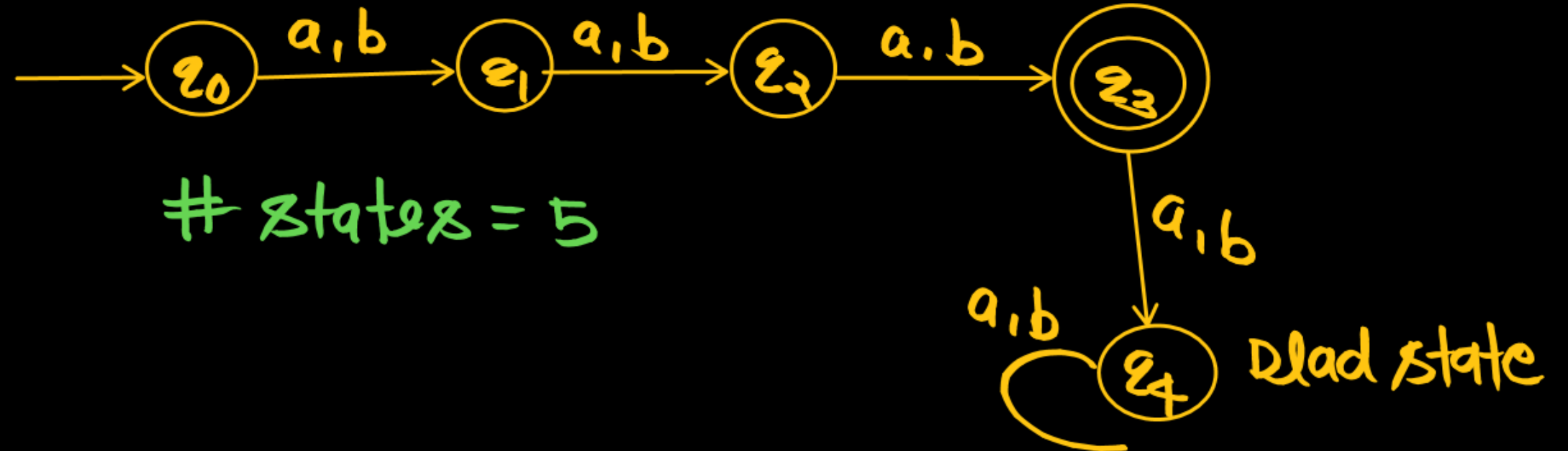
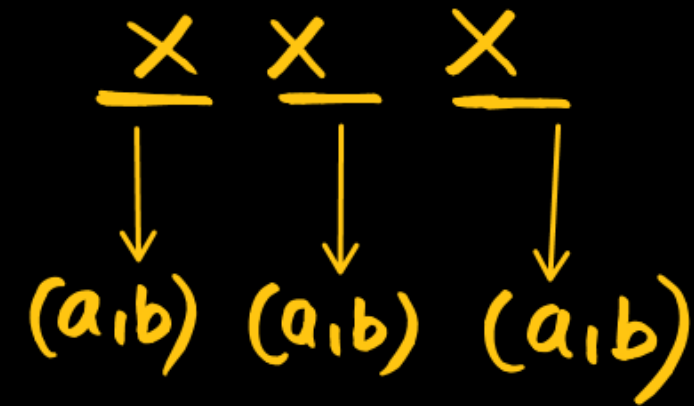
Construct the minimal DFA that Accept all the string of a's and b's where length of the string

- i) Exactly 3
- ii) Atmost 3
- iii) Atleast 3

Soln:

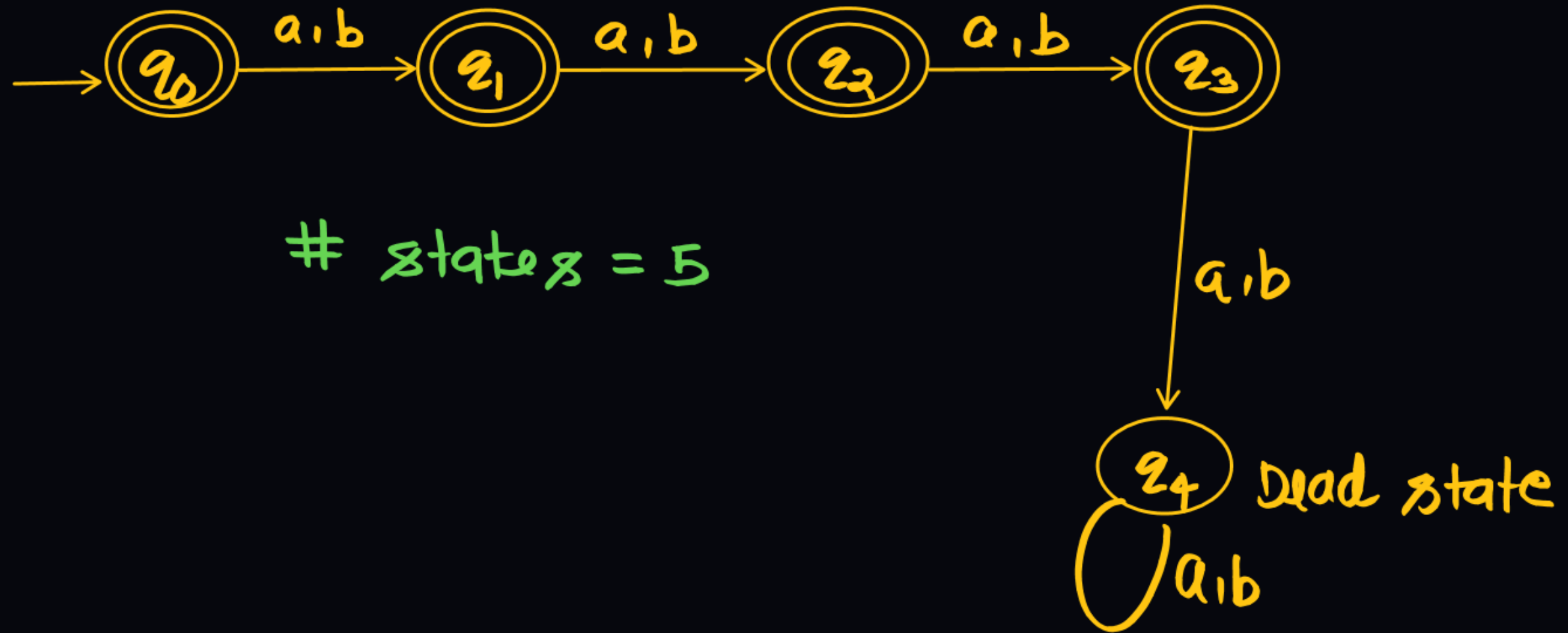
$$(i) \Sigma = \{a, b\}$$

$$|w| = 3$$

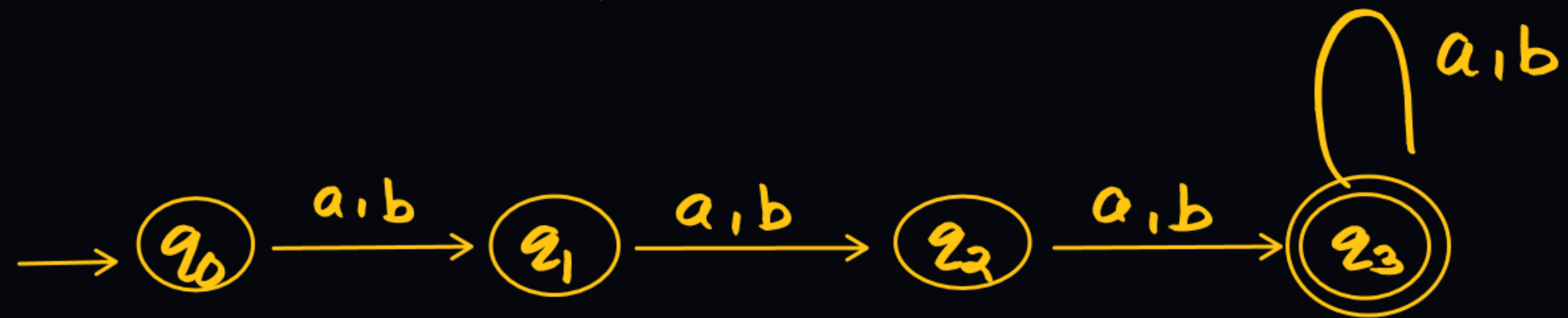


states = 5

(ii) $|w| \leq 3$
 $\downarrow$
0 1 2 3



(iii) $|w| \geq 3$
 $\downarrow$
 3, 4, 5, 6, ...



states = 4

ADRULE

1. $|w| = n \longrightarrow n+2$ states
2. $|w| \leq n \longrightarrow n+2$ states
3. $|w| \geq n \longrightarrow n+1$ states

Question

Construct the minimum DFA that accept all the string of a's and b's where

i) Length of string divisible by 3

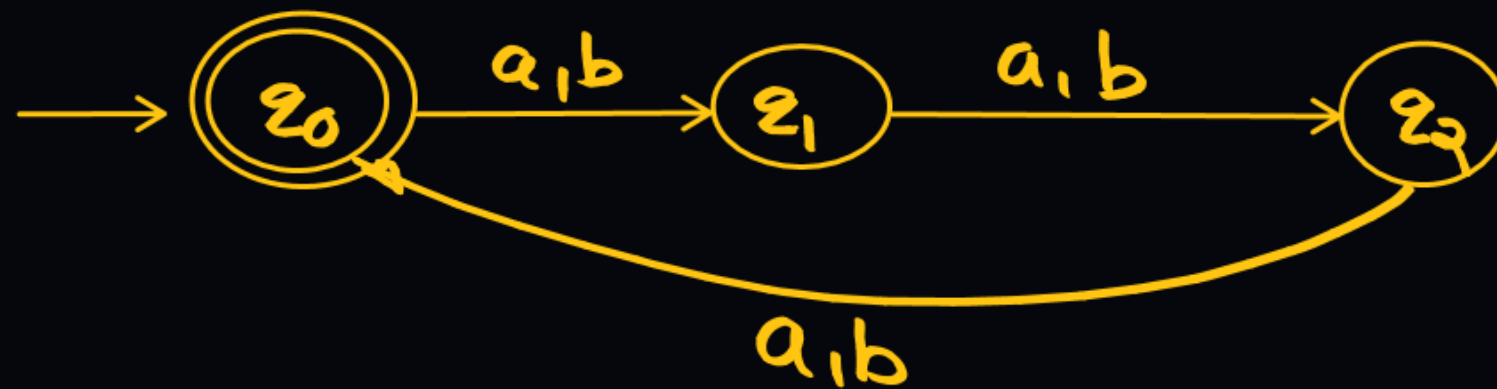
ii) $|w| \equiv 2 \pmod{4}$

iii) $|w| \equiv 3 \pmod{5}$ H.W

iv) $|w| \equiv 2 \pmod{6}$ H.W

v) $|w| \equiv 4 \pmod{7}$ H.W

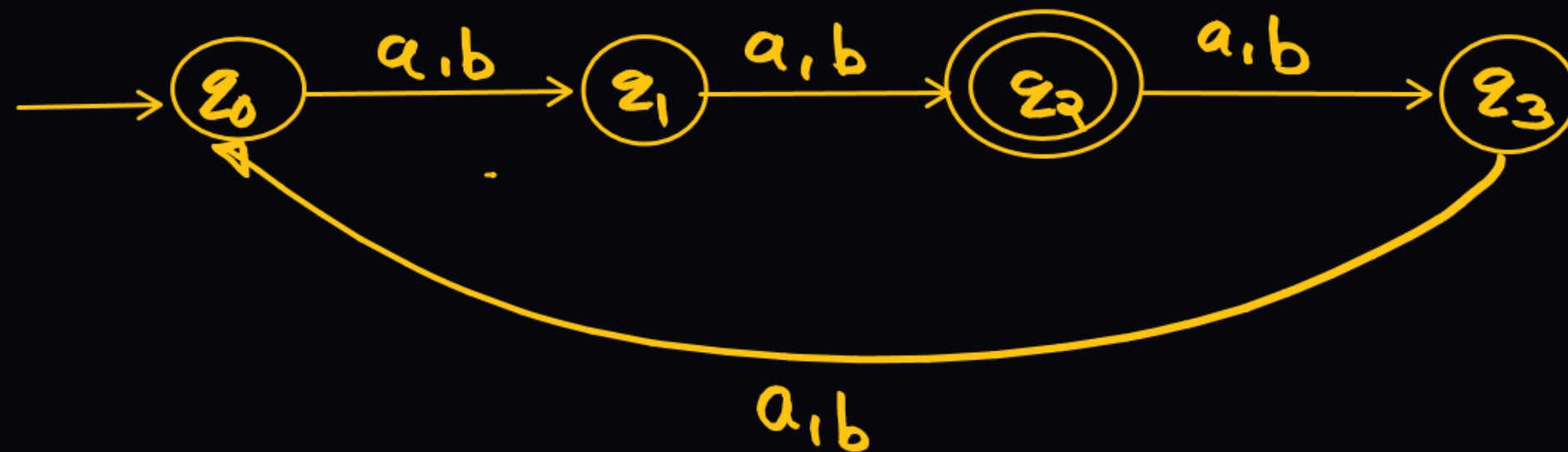
Soln: (i) $|w| = 0 \pmod{3} = 0, 3, 6, 9, 12$



states = 3

(ii) $|w| \equiv 2 \pmod{4} = 2, 6, 10, 14, 18$

↓ Rem
0 1 2 3
 q_0 q_1 q_2 q_3



states = 4

NOTE

ADRULE

Let L is the set of all string of a's and b's where $|w| = r \pmod n$. Here ' r ' represent the remainder and q_r represent the final state. number of states in minimal DFA to accept this language is n .



The smallest finite automaton which accepts the language $\{x \mid \text{length of } x \text{ is divisible by } 3\}$ has

- ☐ A 2 states
- ☒ B 3 states
- ☐ C 4 states
- ☐ D 5 states

Length of string div by 3

$$|w| = 0 \pmod{3}$$

↓
3 state



Question

V.V.Imp

Construct the minimal DFA that accept all the string of 0's and 1's where every string

- 1) ends with 0 $\longrightarrow 1+1 = 2$ state
- 2) ends with 10 $\longrightarrow 2+1 = 3$ state
- 3) ends with 00 $\longrightarrow 2+1 = 3$ state
- 4) ends with 011 $\longrightarrow 3+1 = 4$ state
- 5) ends with 101 $\longrightarrow 3+1 = 4$ state
- 6) ends with 0111 $\longrightarrow 4+1 = 5$ state

1.

$$\Sigma = \{0, 1\}$$

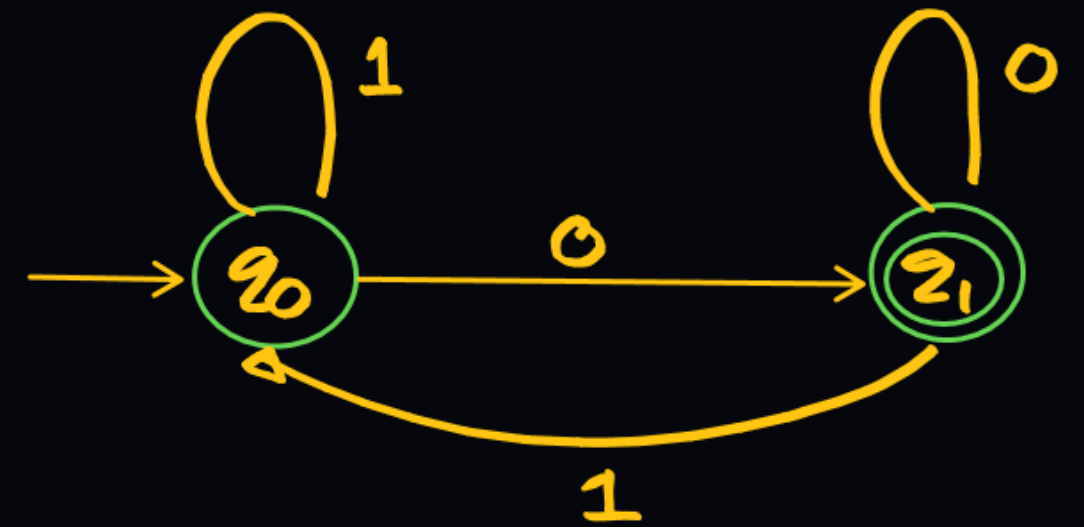
every string ends with 0

$$L = X0$$

$$L = \{ 0, 00, 000, 0000, \dots \}$$

$$10, 110, 1110, \dots$$

$$010, 0110, 01110, \dots$$



states = 2

0 1 0
↑
q1

2. ends with 10

$\frac{10}{A} \frac{0}{R} \frac{10}{A}$

$$L = X10$$

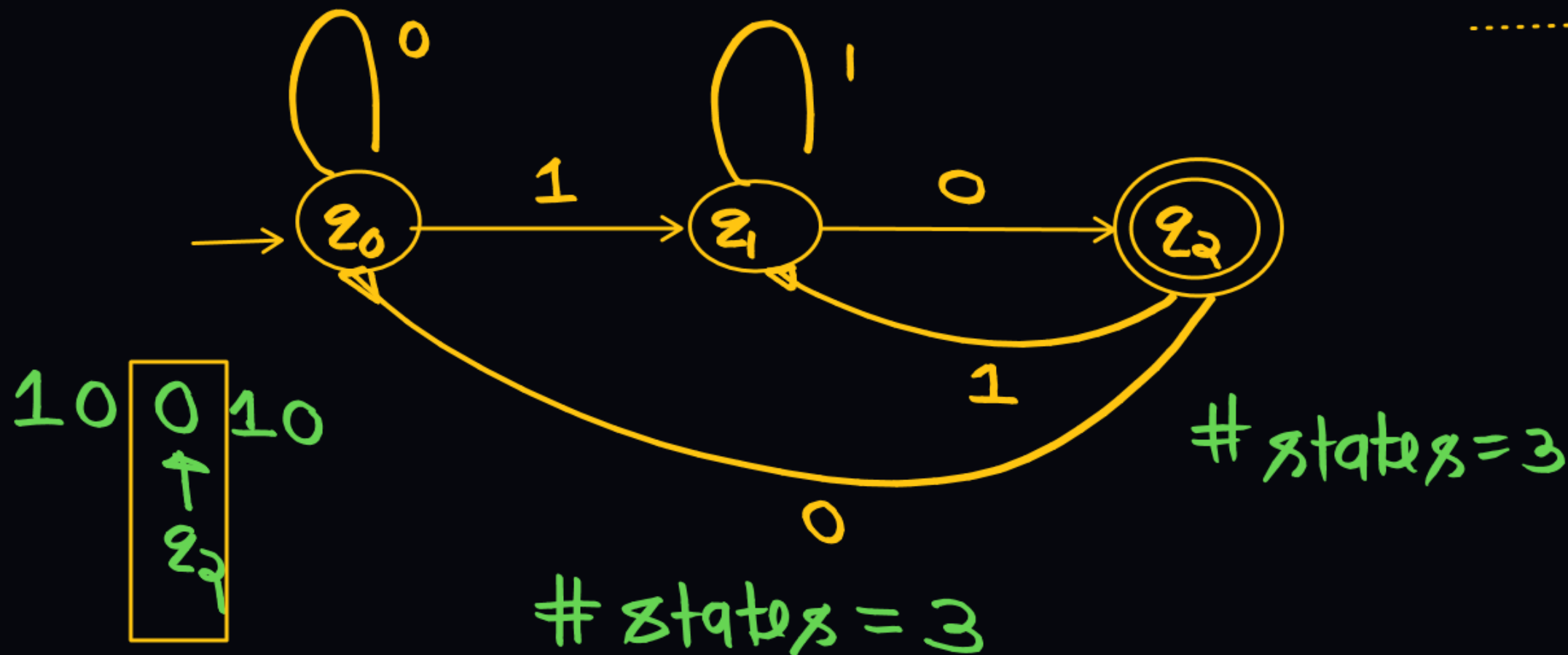
$$L = \{ 10, 010, 0010, 00010, \dots$$

$$110, 1110, 11110$$

$$1010, 101010,$$

..... 00000 10

..... 11111 10



10 1 0
↑
q2

3. ends with 00

$$L = X00$$

$$L = \{ 00, 000, 0000, 00000, \dots \\ 100, 1100, 11100 \}$$

00

1
↑
2 ₂

 00

..... 111 00



states = 3

0

1
↑
2 ₁

 00

4. ends with 011

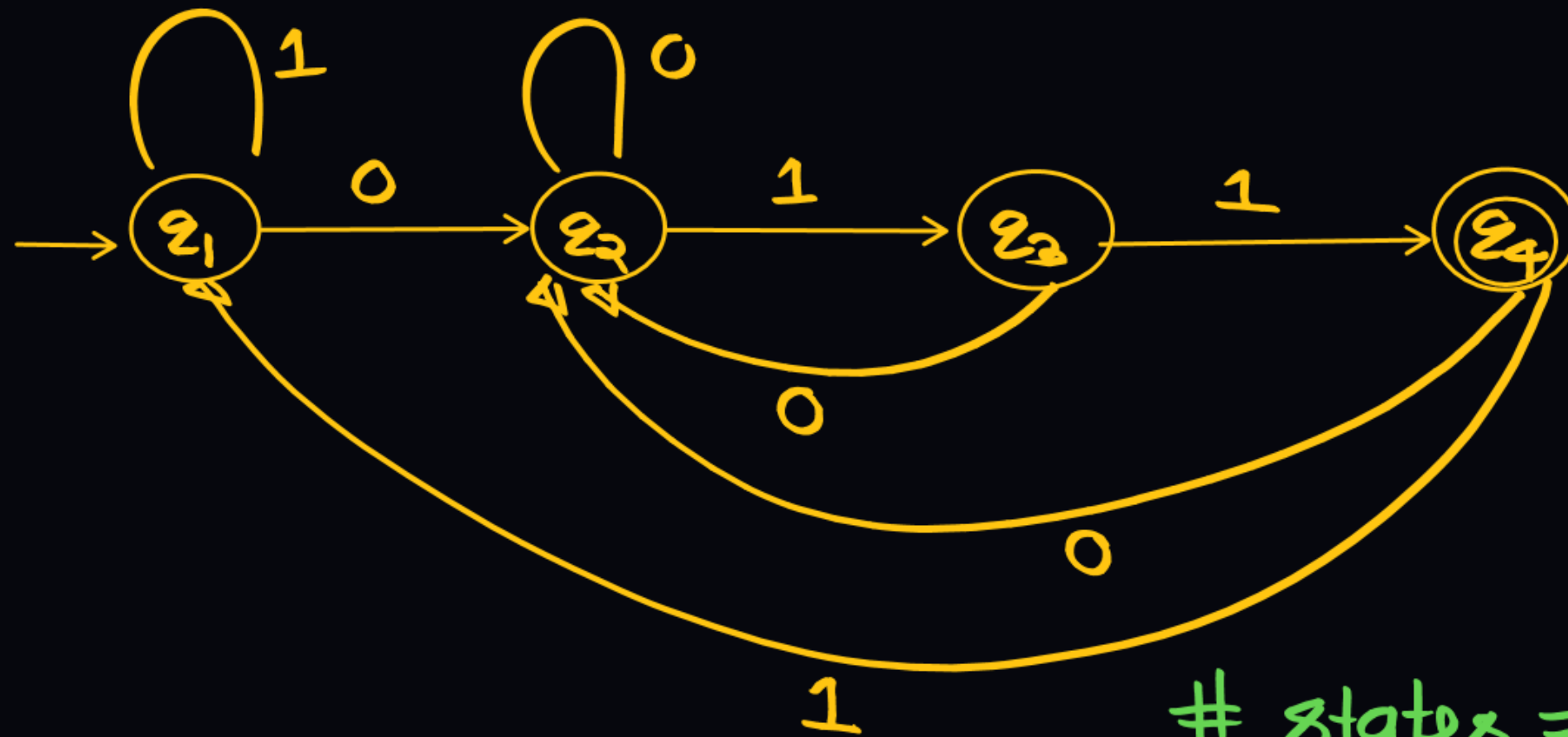
$$w = x011$$

$$L = \{011, 1011, 11011\}$$

01011
↑
23

011011
↑
24

0111011
↑
24



states = 4

5. ends with 101

$$L = X 101$$

$$L = \{ 101, 0|01, 00|01, 1101, 11101 \}$$



states = 4

10 0
↑
q₂ 101

101 0
↑
q₃ 1

101 1
↑
q₃ 01

NOTE

ADRULE

Let L is the set of all string of 0's and 1's where every string ends with a substring S of length n i.e $L=XS$, $|s|=n$. Then number of states in minimal DFA to accept this language is $n+1$



Question

V.V. Imp



Construct the minimal DFA that accept all the string of 0's and 1's where

- A) 2nd symbol from right ends is 1
- B) 3rd symbol from right end is 1
- C) 2nd symbol from right end is 0
- D) 3rd symbol from right end is 0

Soln: and symbol From Right end is 1

$L = \dots 1 X$
 $\downarrow$
 $(0, 1)$

$L = \{ 10, 11, 111, 1111, \dots$
 $110, 1110,$
 $010, 0010$

10 1
↑
 z_2 0

10 0
↑
 z_3 10



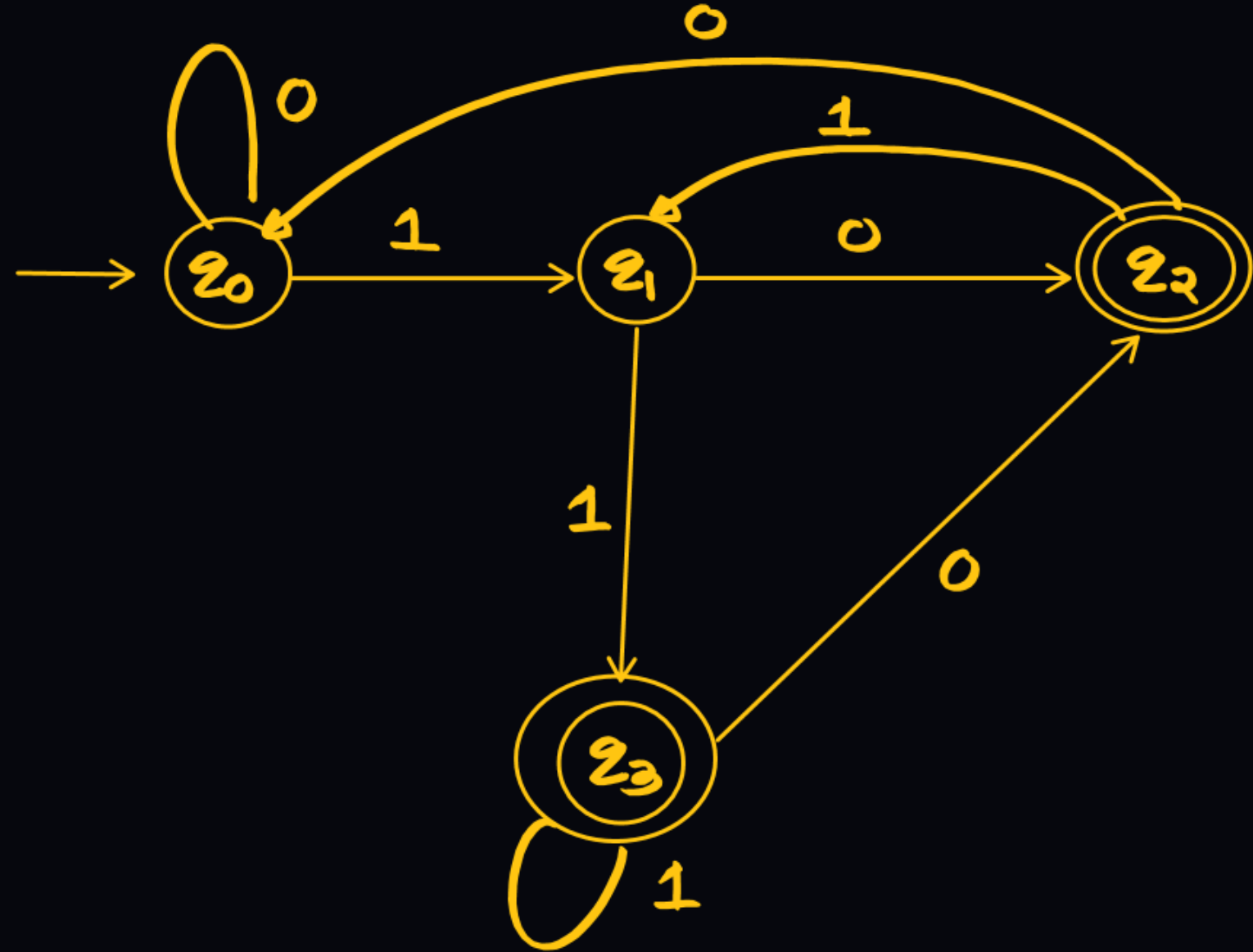
states = 4

ADRULE

δ	0	1
q_0	q_0	q_1
q_1	q_2	q_3
q_2	q_0	q_1
q_3	q_2	q_3

Fixed

Odd No. of states



2. 3rd symbol From Right end is 1

$L = \dots\dots 1 X X$

$\downarrow \quad \downarrow$
 $(0,1) (0,1)$
 Fixed

	0	1
$\rightarrow q_0$	q_0	q_1
q_1	q_2	q_3
q_2	q_4	q_5
q_3	q_6	q_7
q_4	q_0	q_1
q_5	q_2	q_3
q_6	q_4	q_5
q_7	q_6	q_7

odd

(i) $w = 00100(A)$

$(q_0, 00100)$

$q_0, 0100$

$q_0, 100$

$q_1, 00$

$q_2, 0$

q_4

Accepted

(ii) $w = 101011(R)$

$(q_0, 101011)$

$q_1, 01011$

$q_2, 1011$

$q_5, 011$

$q_2, 11$

$q_5, 1$

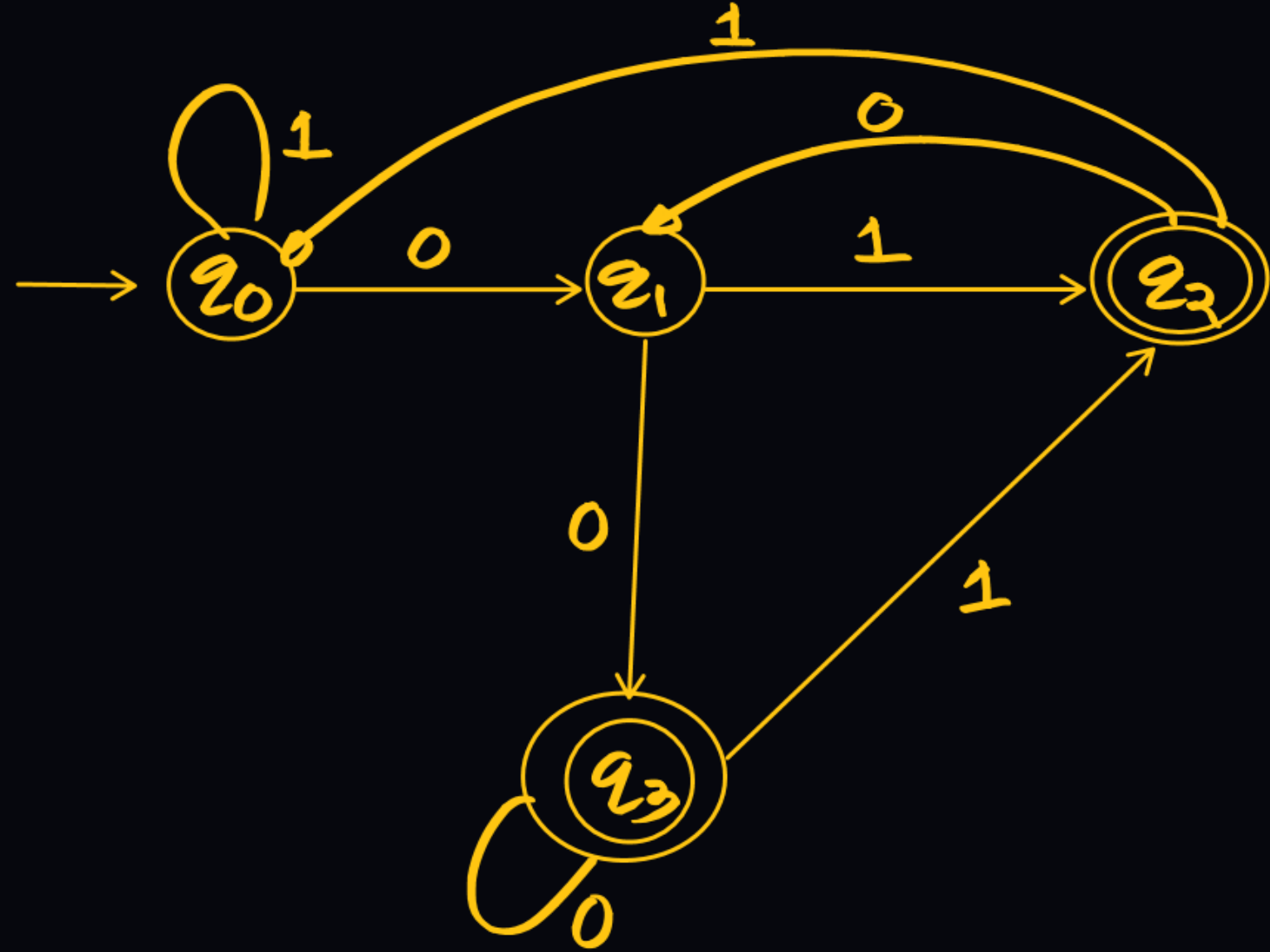
q_3

[Rejected]
 Non Final

ADRULE (3) 2nd symbol From Right end is 0

→ Fixed

δ	0	1
q_0	q_1	q_0
q_1	q_3	q_2
q_2	q_1	q_0
q_3	q_3	q_2



4. 3rd symbol From Right end is 0

$L = \dots 0XX$
 $\downarrow \quad \downarrow$
 $(0,1) (0,1)$

	$\rightarrow$ Fixed	
	0	1
$\rightarrow z_0$	z_1	z_0
z_1	z_3	z_2
z_2	z_5	z_4
z_3	z_7	z_6
z_4	z_1	z_0
z_5	z_3	z_2
z_6	z_5	z_4
z_7	z_7	z_6

NOTE

ADRULE

- Let L is the set of all string of 0's and 1's where n th symbol from Right end is fixed . Then number of states in minimal DFA to accept this language is 2^n
- 2^{n-1} are Final state(last 50 % are final state)
- Odd number of states are under fixed symbol



Question

V.V.Imp

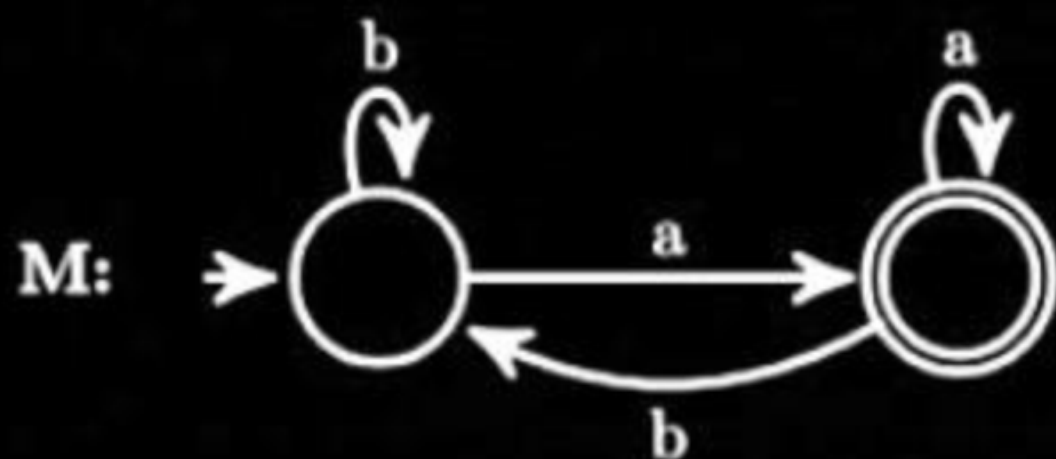


Construct the minimal DFA that accept all the string of 0's and 1's where every string contain the substring

- 1) 0
- 2) 10
- 3) 010
- 4) 011
- 5) 0110

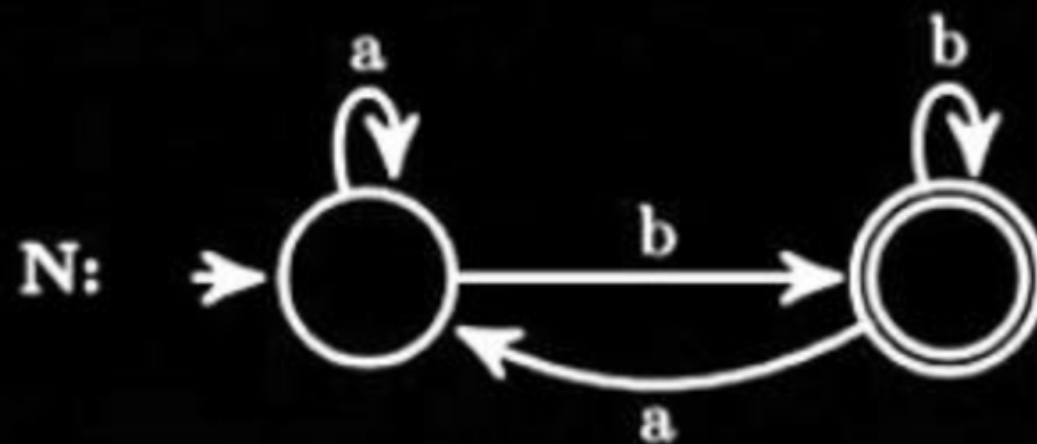


Consider the DFAs M and N given above. The number of states in a minimal DFA that accept the language $L(M) \cap L(N)$ is _____. **Ans: 1**

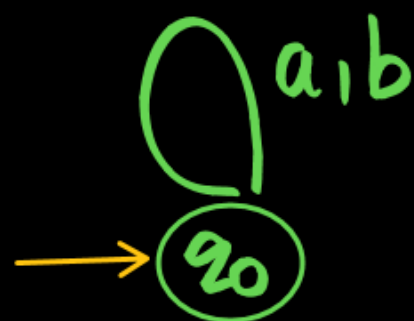


ends with a

$$L(M) \cap L(N) = \phi$$



ends with b



$L = \phi$ [Minimal DFA = 1 state]

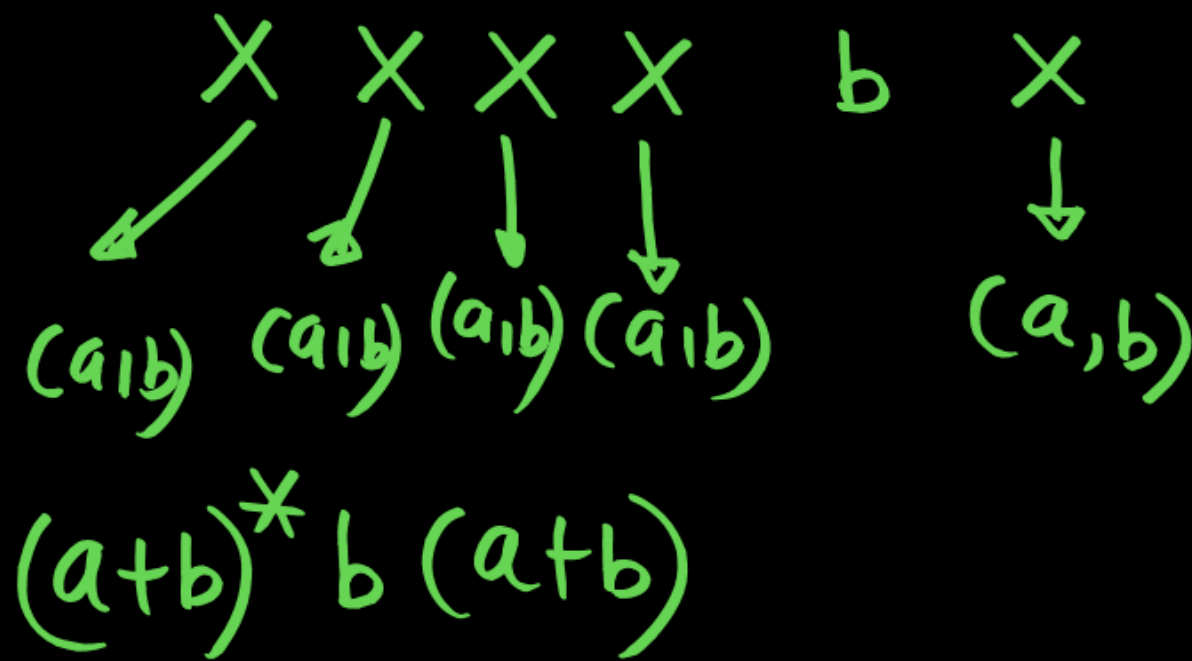


The number of states in the minimal deterministic finite automaton corresponding to the regular expression $(0 + 1)^*(10)$ is _____.

↓
ends with 10

$$\begin{aligned}\text{No. of states in MDFA} &= n+1 \\ &= 2+1 = 3\end{aligned}$$

Consider the language L given by the regular expression $(a + b)^*b(a + b)$ over the alphabet $\{a, b\}$. The smallest number of states needed in a deterministic finite-state automaton (DFA) accepting L is _____.



2nd symbol From Right end
is b

$$\begin{aligned}
 \text{No. of states in MDFA} &= 2^n \\
 &= 2^2 \\
 &= 4
 \end{aligned}$$



Thank You